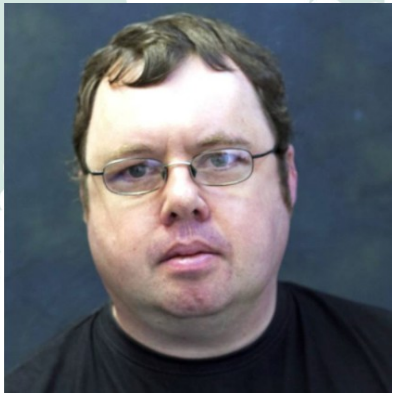
UNIVERSITY OF PLYMOUTH

Welcome to MSc Data Science and Business Analytics

Induction Thursday 24th September 2026



Welcome to the MSc Data Science and Business Analytics!!!



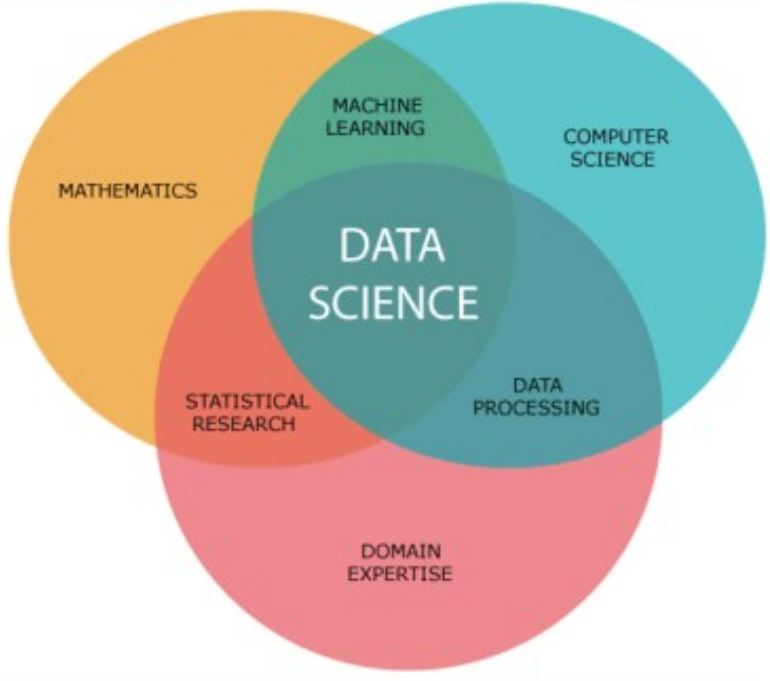
- *Craig McNiele* Associate Professor of Theoretical Physics
- Programme Manager of the MSc Data Science & Business Analytics Health Data Science and Statistics MSc

- 2 Kirkby Place, ground floor, room 004

craig.mcneile@plymouth.ac.uk



Data Science and Business Analytics



Business analytics refers to the statistical methods and computing technologies for processing, mining and visualizing data to uncover patterns, relationships and insights that enable better business decision-making.

Some examples

- Data Challenge Cornwall

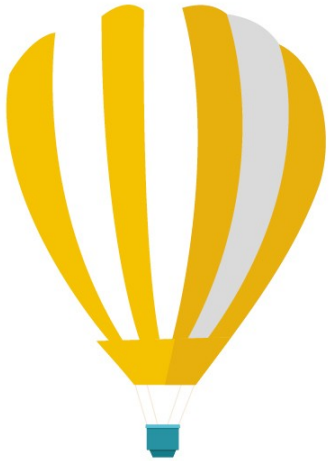
  Videos at Data Challenge Cornwall

  - ***Visualising GP Surgery Data***
  - ***Crowdfunder, Realtime Data Visualisation of Pledge Activity and Trends***
  - ***Coastline Housing, Predicting Faults & Identifying Resident Support Needs***
  - ***Cornish Lithium, Optical Character Recognition of Historical Mining Records***



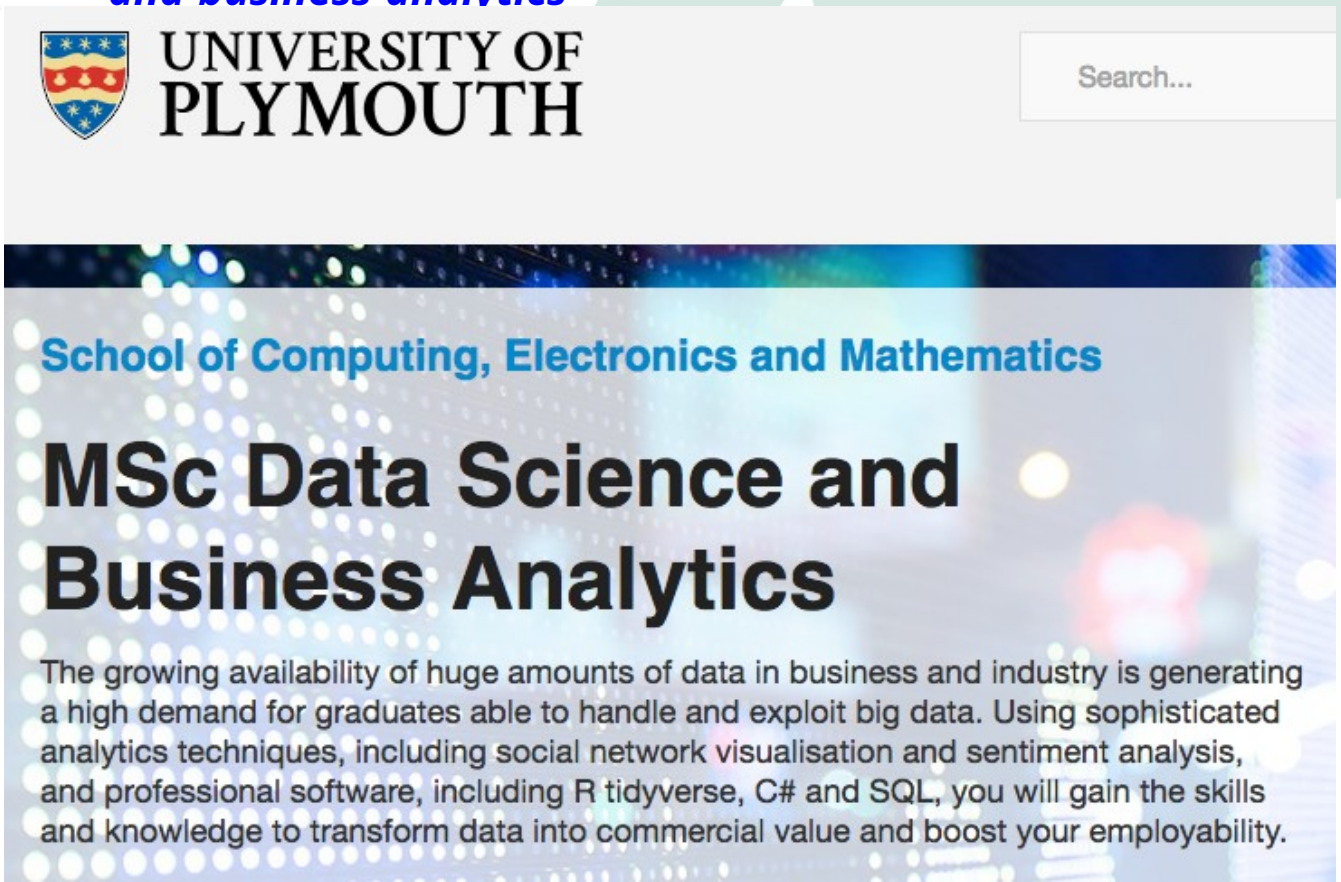
Course Overview

- Purpose
  - ***Using sophisticated analytics techniques, including social network visualization and sentiment analysis, and professional software, including R tidyverse, Python and SQL databases, you will gain the skills and knowledge to transform data into commercial value and boost your employability.***



The course web site

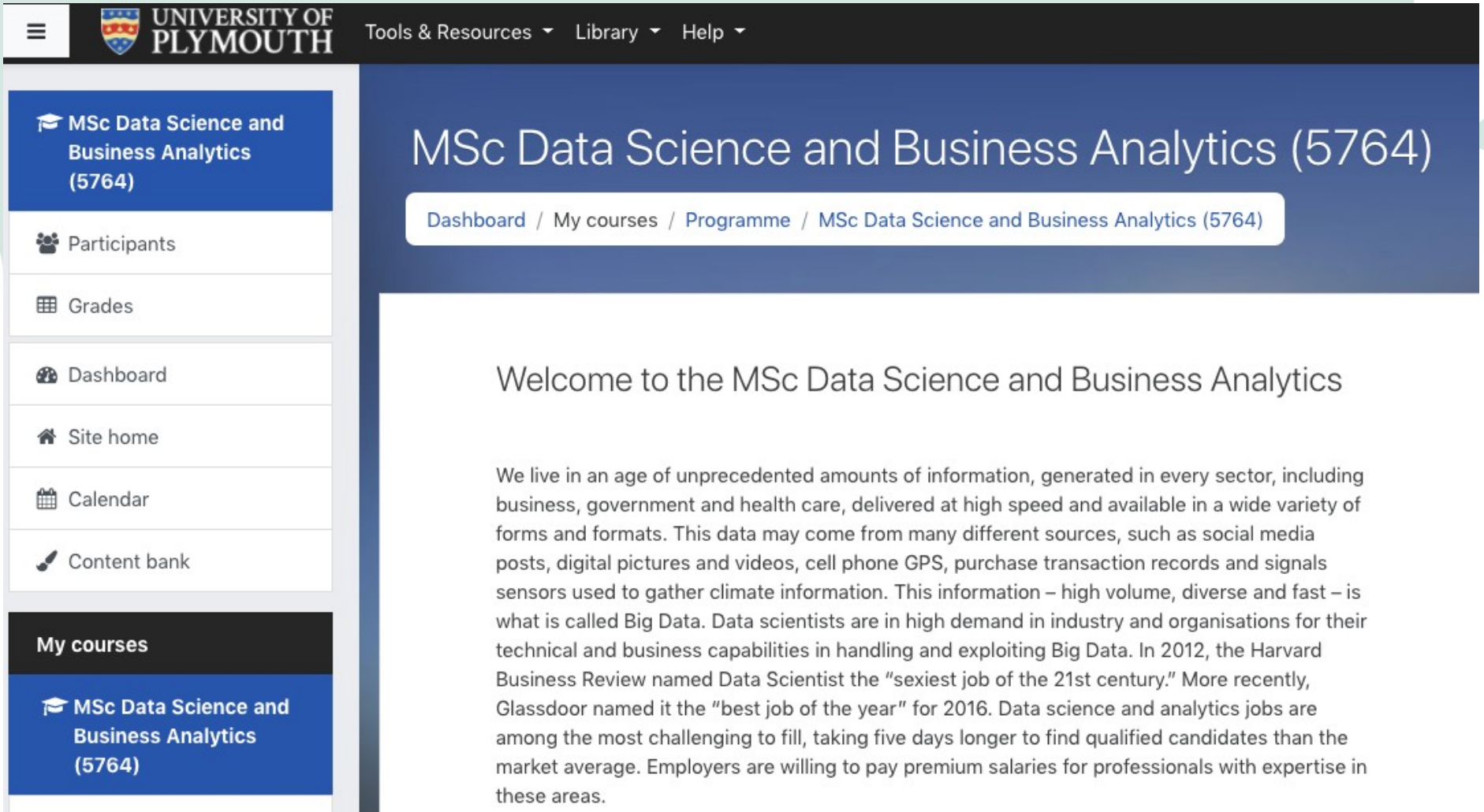
- ***https://www.plymouth.ac.uk/courses/postgraduate/msc-data-science-and-business-analytics***





The course DLE site

- ***https://dle.plymouth.ac.uk/course/view.php?id=24362***

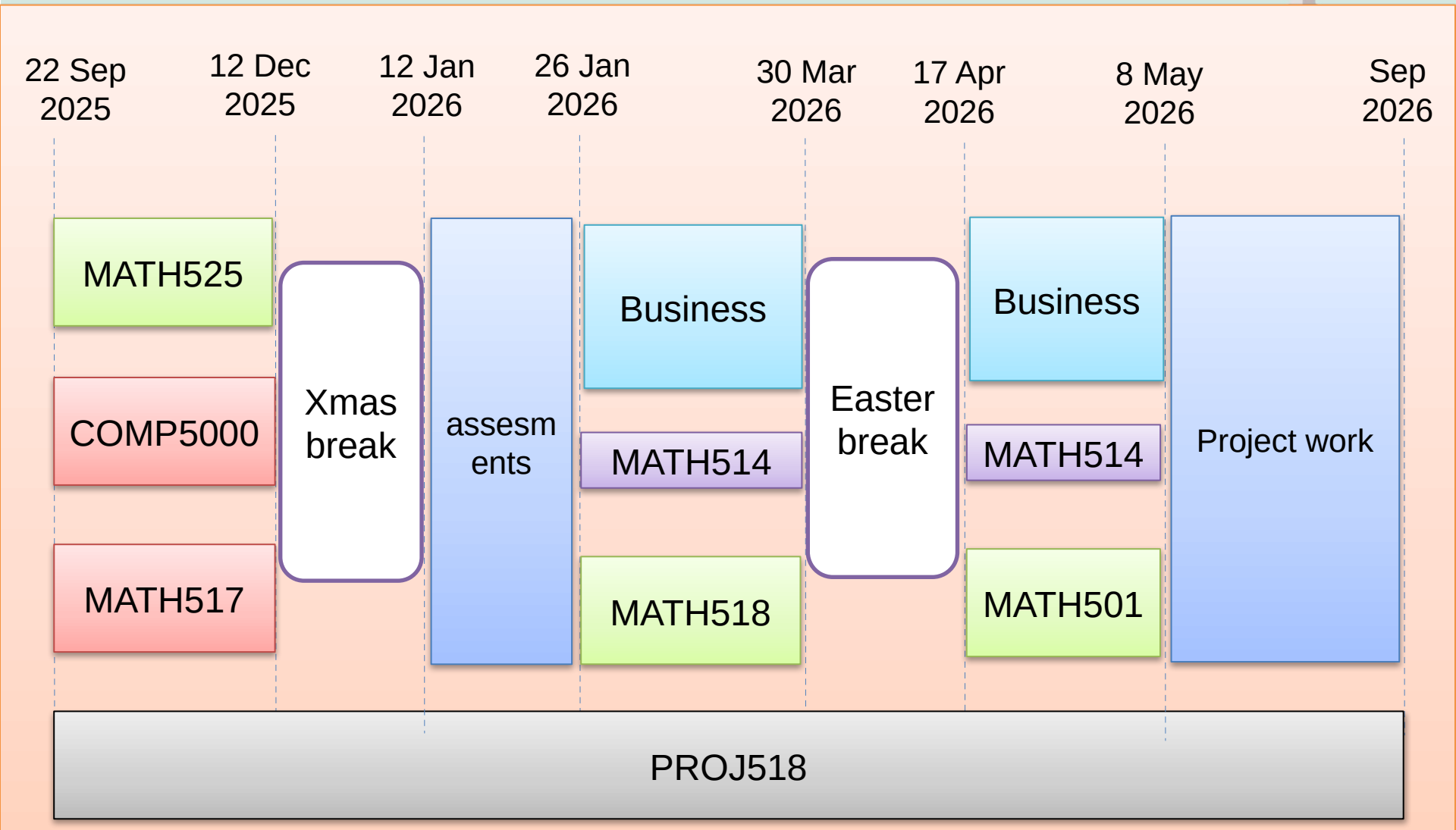


DSBA Course Overview

- Semester 1
  - ***MATH517 Big Data Visualization and Analytics***
  - ***MATH525 Business Analytics***
  - ***COMP5000 Software Development and Databases***
- Semester 2
  - ***MATH514 Contemporary issues in data science***
  - ***MATH518 Applied Data Modelling and Artificial Intelligence***
  - ***Optional Business Module***
- All Year
  - ***PROJ518 MSc Dissertation and Research Skills***



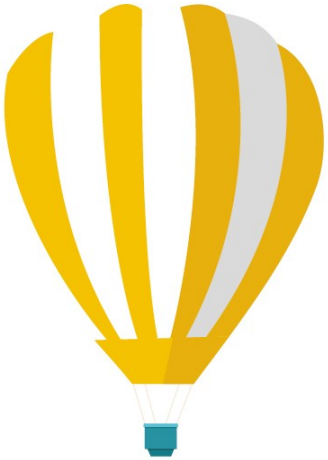
DSBA Course Calendar



MATH513 Big Data Visualisation and Analytics

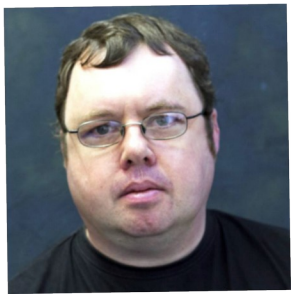
- Module Staff
  - ***Vincent Drach***
- Learning Objectives (100% coursework)
  - ***Provide experience of data visualization***
  - ***Develop a working knowledge of R using RStudio and an opportunity for to produce a professional analysis report***





COMP5000 Software Development and Databases

- Module Staff
  - ***Craig McNeile***
- Learning Objectives (100% coursework)
  - ***Implement computer programs and databases***
  - ***Python, R, SQL***
  - ***Introduce students to the relational model, object-orientation and event-driven interactive development***





MATH525 Business Analytics

- Module Staff
- **Nathan Broomhead**
- Learning Objectives (100% coursework)
  - ***To provide an understanding of the key challenges of advanced data management and analysis in a financial and business management context.***
  - ***To offer an appreciation of state-of-the-art business analytics and technologies that can support efficient and effective management decisions to complex problems in financial and business applications.***
  - ***To model and demonstrate solutions for decision-making problems in different business settings.***



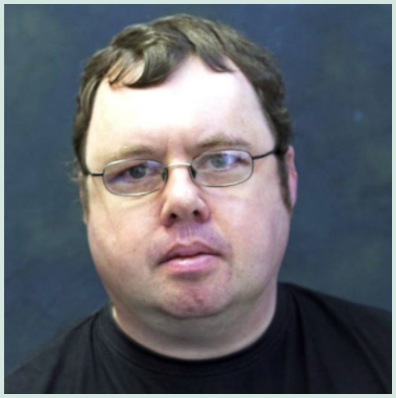
MATH518 Applied Data Modelling and Artificial Intelligence

- Module Staff
  - ***Julian Stander***
- Learning Objectives (100% coursework)
  - ***Machine Learning***
  - ***Provide students with broad tool boxes for frequentist and Bayesian modelling***
  - ***Employment-relevant problem solving experience.***





MATH514 Contemporary Issues in Data Science



- Module Staff
  - ***Guest lecturers***
  - ***Craig McNeile***
- Learning Objectives (100% coursework)
  - ***This module utilizes acknowledged experts as guest lecturers to teach about 'hot topics' in their field of data science.***

Example talks from 2024

- ***Sri Harish Kalidass (BT Research) Leveraging AI for Future Cyber Defence Research in BT***
- ***Gareth Tucker (KPMG) KPMG and data***

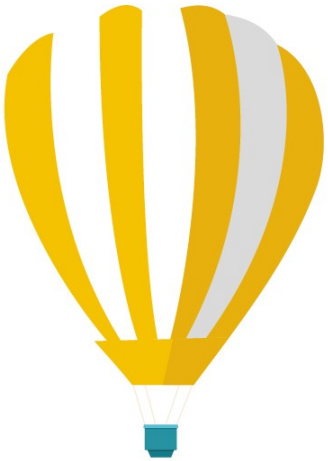
Business Optional Module

- Choose 1 module from
  - ***Finance:***
    - ACF7003 International Finance
    - ACF7004 Analysis of Financial Statements (*ML:*
  - ***Marketing:***
    - MKT7004 Social Media Theory and Practice (*ML: Andrew Morton*)
  - ***International Business:***
    - STO7002 Current Issues in International Business (*ML: Atul Mishra*)
    - STO7012 International Business Strategy for Competitive Advantage (*ML: Charles Thornton*)
    - STO7006 Operations and Project Management (*ML: Stephen Childe*)

For more information on the Business Optional Modules, please contact the relevant Module Leaders
https://www.plymouth.ac.uk/courses/postgraduate/msc-data-science-and-business-analytics

How to choose the Business Optional Modules?

- Details to be sent out by the faculty office soon.
- Online module choice for full time students
- Different system for part time students



PROJ518 MSc Dissertation and Research Skills

- Module Leader
  - ***Martyn Hann***
  - ***Supervisors***
- 60 Credits
- All year
- Most of the project is done May to September
- List of projects provided by staff
- Develop your own idea



What if there is a problem?

- Lecturers/Module leaders
- Programme leader
  - ***craig.mcneile@plymouth.ac.uk***
  - ***2 Kirkby Place, ground floor, room 004***
- Personal tutor: Craig McNeile
- The Student Union
  - ***https://www.upsu.com/***



Student Representative(s)

- You need to select one or two representatives
  - ***Contact point***
  - ***Committee member***
- More information on the role of student representative is available on the course DLE webpage, under “Course Rep Information”.
- If you are interested and would like to nominate yourself as Course Rep, please email me by **Monday the 5th October** with a short paragraph explaining why students should vote for you.

